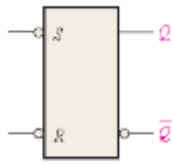
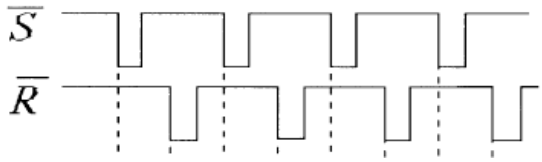
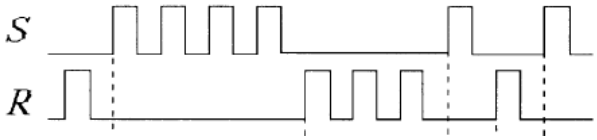
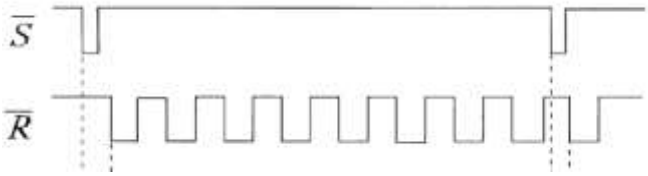
PROBLEM SET 9

1. Obtain the output waveform from SR latch for the given inputs

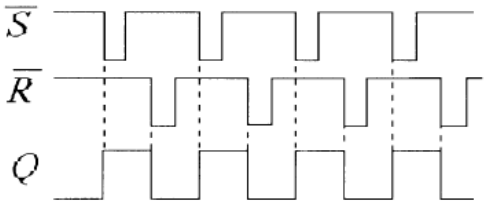


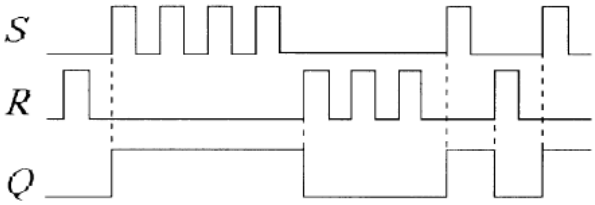
a.


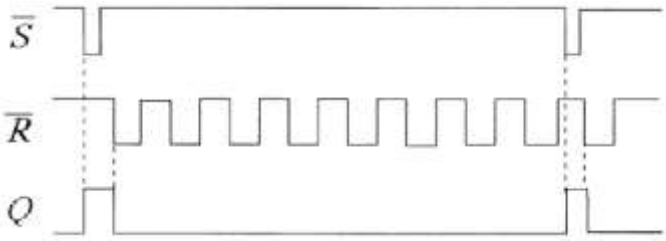
b.


c.


Solution:

a.


b.


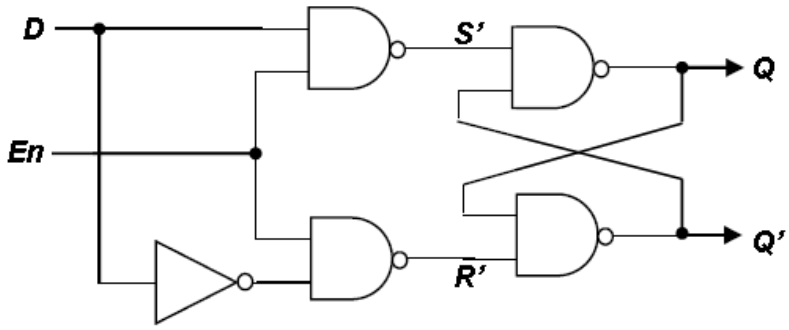
c.


2. Problem 5.1 From Digital Circuits 5th edition Morris Mano.

(**Problem 5.1**)

*Inputs for SR Latch using NAND gates are **S'** and **R'**:*

***S** = **D En** ⇒ **S'** = (**D En**)'*
***R** = **D' En** ⇒ **R'** = (**D' En**)'*



***(a)** NOR gates for the SR Latch part and AND gates for the other two:*
Inputs for SR Latch using NOR gates are ***S*** and ***R***:
⇒ ***S** = **D En*** and ***R** = **D' En*** are directly given by AND gates.

***(b)** NOR gates for all four gates:*
***S** = **D En** = (**D'** + **En'**)'* and ***R** = **D' En** = (**D** + **En'**)'* can be obtained using NOR gates.
One inverter will be necessary for generating ***En'***.

***(c)** Four NAND gates without an inverter:*
Instead of using an inverter for obtaining ***D'*** required for generating ***R'*** = (***D' En***)',
generate ***R'*** by the logic ***R'*** = (***S' En***)' = ((***D En***)'***En***)' = ((***D'*** + ***En'***)***En***)' = (***D' En***)'.

2. Problem 5.6

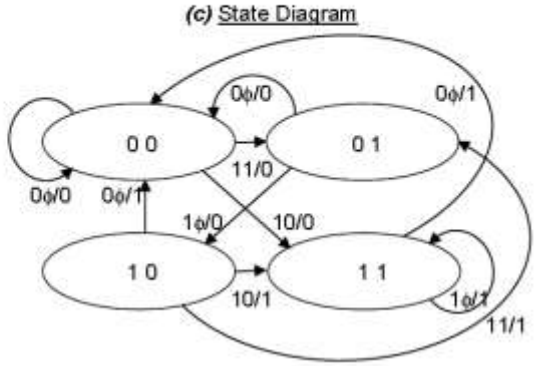
(**Problem 5.6**) *A sequential circuit with two D flip-flops A and B, two Inputs x y and one Output z is specified by the State and Output equations A(t+1) = xy' + xB, B(t+1) = xA + xB', z = A.*

***(a)** Logic diagram:* The D inputs of the two flip-flops are given by $D_A = xy' + xB$, $D_B = xA + xB'$.

(b) State Table

A(t)	B(t)	A(t + 1) B(t + 1)			
		x = 0 y = 0	x = 0 y = 1	x = 1 y = 0	x = 1 y = 1
0	0	0 0	0 0	1 1	0 1
0	1	0 0	0 0	1 0	1 0
1	0	0 0	0 0	1 1	0 1
1	1	0 0	0 0	1 1	1 1

(c) State Diagram



3. problem 5.7,

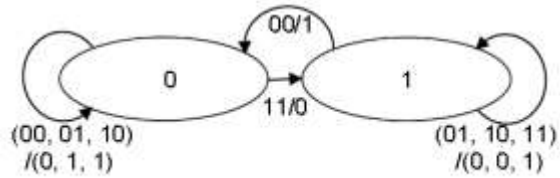
(**Problem 5.7**) *One D flip-flop Q; two inputs x, y; Full Adder output C S = Q plus x plus y; D = C.*

$C = x'yQ + xy'Q + xyQ' + xyQ$ and $S = x'y'Q + x'yQ' + xy'Q' + xyQ$.
State given by Q. State Equation: $Q(t + 1) = C(t)$. Output = $S(t)$.

State Table

$Q(t)$	$Q(t+1)$			
	$x=0$ $y=0$	$x=0$ $y=1$	$x=1$ $y=0$	$x=1$ $y=1$
0	0	0	0	1
1	0	1	1	1

State Diagram



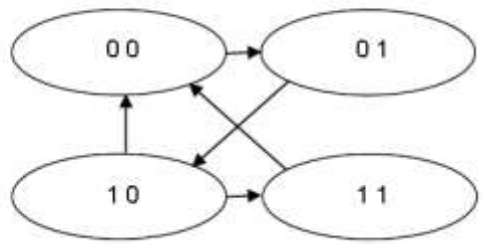
4. Problem 5.8,

(**Problem 5.8**) *Two T flip-flops A and B;* $T_A = A + B$, $T_B = A' + B$.

State Table

$A(t)$	$B(t)$	$A(t+1)$	$B(t+1)$
0	0	0	1
0	1	1	0
1	0	0	0
1	1	0	0

State Diagram



5. Problem 5.9,

(**Problem 5.9**) *Two JK flip-flops A and B; one input x;* $J_A = x$, $K_A = B$, $J_B = x$, $K_B = A'$.

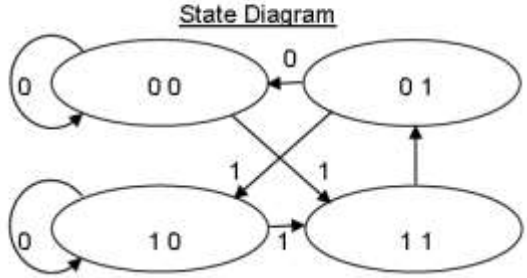
(a) *State Equations:*

$A(t + 1) = J_A A' + K_A' A = xA' + AB'$ and $B(t + 1) = J_B B' + K_B' B = xB' + AB$.

(b) State Table

$A(t)$	$B(t)$	$A(t+1)$ $B(t+1)$	
		$x=0$	$x=1$
0	0	0	1
0	1	0	1
1	0	1	1
1	1	0	0

State Diagram



6. Problem 5.10

(**Problem 5.10**) *Two JK flip-flops A and B; two inputs x and y; one output z.*

$J_A = Bx + B'y'$, $K_A = B'xy'$, $J_B = A'x$, $K_B = A + xy'$; $z = Ax'y' + Bx'y'$.

(b) *State Table:*

A(t)	***B(t)***	***A(t + 1) B(t + 1)***				
			x = 0 ***y*** = 0	***x*** = 0 ***y*** = 1	***x*** = 1 ***y*** = 0	***x*** = 1 ***y*** = 1
0	0	***y'*** , ***x***	1 0	0 0	1 1	0 1
0	1	***x*** , ***x'+y***	0 1	0 1	1 0	1 1
1	0	***x'+y*** , ***0***	1 0	1 0	0 0	1 0
1	1	1 , 0	1 0	1 0	1 0	1 0